A Multitask Deep Learning Model for Classification and Regression of Hyperspectral Images: Application to a Large-Scale Dataset

Koushikey Chhapariya, *Graduate Student Member, IEEE*, Alexandre Benoit, *Member, IEEE*, Krishna Mohan Buddhiraju, and Anil Kumar

***Abstract*— Multitask learning is a widely recognized technique in the field of computer vision and deep learning domain. However, it is still a research question in remote sensing, particularly for hyperspectral imaging (HSI). Moreover, most of the research in the remote sensing domain focuses on small and single-task-based annotated datasets, which limits the generalizability and scalability of the developed models to more diverse and complex real-world scenarios. Thus, in this study, we propose a multitask deep learning model designed to perform multiple classification and regression tasks simultaneously on hyperspectral images. We validated our approach on a large hyperspectral dataset called TAIGA, which contains 13 forest variables, including three categorical variables and ten continuous variables with different biophysical parameters. We design a sharing encoder and task-specific decoder network to streamline feature learning while allowing each task-specific decoder to focus on the unique aspects of its respective task. In addition, a dense atrous pyramid pooling layer and attention network were integrated to extract multiscale contextual information and enable selective information processing by prioritizing task-specific features. Furthermore, we computed multitask loss and optimized its parameters for the proposed framework to improve the model performance and efficiency across diverse tasks. A comprehensive qualitative and quantitative analysis of the results shows that the proposed method significantly outperforms other state-of-the-art methods. We trained our model across ten seeds/trials to ensure robustness. Our proposed model demonstrates higher mean performance while maintaining lower or equivalent variability. To make the work reproducible, the codes will be available at https://github.com/Koushikey4596/Multitask-Deep-Learning-Model-for-Taiga-datatset.**

***Index Terms*— Deep learning, hyperspectral image classification, multitask, spectral–spatial, TAIGA dataset.**

Received 18 July 2024; revised 6 December 2024 and 25 January 2025; accepted 3 March 2025. Date of publication 24 March 2025; date of current version 17 April 2025. *(Corresponding author: Koushikey Chhapariya.)*

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Digital Object Identifier 10.1109/TGRS.2025.3553751

I. Introduction

HYPERSPECTRAL imaging (HSI) is a remote sensing technique (also known as image spectrometry) consisting of hundreds of contiguous and spectrally narrow wavelength bands. This detailed spectral information helps in distinguishing similar features and has applications in various domains, such as agriculture sector [1], [2], urban target detection [3], [4], mining industry [5], [6], environmental and natural resource management [7], [8], and surveillance [9], [10]. However, the high dimensionality of data increases storage and computational costs. Moreover, the curse of dimensionality [11] and redundancy in adjacent spectral bands limit the accuracy of HSI classification. Thus, there is a need for automated learning algorithms to reduce the time consumption for manual work and increase overall efficiency. Many machine learning algorithms, such as KNN, SVM, and MLP, have performed satisfactorily but require experts to manually tune parameters that fail to handle unknown/unseen and complex scenes. This increases the need for more robust and adaptive learning methods, thus introducing deep learning algorithms with HSI [12], [13]. Convolutional neural networks (CNNs) are one such deep learning approach that is widely considered for extracting learned features with its hierarchical properties and layered structure.

The acquired hyperspectral images can be represented as a data cube with two continuous axes of spatial dimensions and one axis for spectral dimension. Several researchers [14], [15], [16] have demonstrated the importance of using both spectral and spatial information for hyperspectral image analysis to improve image classification accuracy. The 3-D CNN algorithm can extract discriminatory features from the HSI data cube considering its spectral and spatial information. Despite several research works on HSI with CNN, there is still a limitation with the availability of hyperspectral datasets. Moreover, the prediction of continuous variables from HSI using CNN is little known and understudied [17], while the majority of studies in the field have focused on land cover classification, typically approached as a single-task method. In addition, learning algorithms demand substantial amounts of data before yielding meaningful results. Thus, computational expenses rise when dealing with multiple

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tasks, each demanding significant computational resources and extensive training [18].

Multitask learning [19] aims at solving multiple prediction problems from a given data sample to improve learning efficiency and classification accuracy by sharing information among different tasks. Multitask learning is a well-known method in the computer vision domain, while it is still a research question in the remote sensing domain [20]. The main motivation for this research is to develop a learning-based algorithm that can enable multitask learning for both regression and classification tasks while maintaining or improving inference speed performances on an independent test set. Despite their distinct objectives, regression and classification tasks in remote sensing share commonalities in supervised learning and are expected to utilize similar learning patterns and techniques. This common ground forms the basis for employing a multitask approach, where a single model is trained to handle both regression and classification tasks simultaneously. While significant research has been conducted on multitask learning models, particularly in the context of image segmentation [21], [22], [23], there has been relatively limited exploration of multitask learning within hyperspectral image classification [24]. With this observation, our research aims to address the following research question.

1) Is there an existing multitask learning framework capable of simultaneously predicting different classes and computing values for continuous variables specifically designed for large hyperspectral datasets?
2) How does multitask behave when applied to hyperspectral images, particularly considering the presence of a large number of bands, and what impact does the integration of spectral and spatial information have on its performance?
3) Are there effective methods available to handle a class imbalance in the data and achieve task balancing to compute the multitask loss and optimize its parameters within a multitask learning framework?

This article proposes a multitask deep learning framework for classification and regression for a large hyperspectral dataset to answer the abovementioned questions. The main focus of our work is to propose a framework that can assign classes to input data and predict continuous outcomes simultaneously. We considered an encoder–decoder-based framework for integrating multiple tasks within a single model and shared representation learning. Furthermore, we append a dilated (atrous) convolutional module to improve the network's contextual reasoning. From a shared latent representation, the multitask decoder is designed, where we first separate the classification and regression tasks into two distinct paths, each being more task-specific. Finally, task-specific heads are optimized, relying on a single fully connected layer. The majority of the model's parameters and depth are in the feature encoding, with very little complexity in each task decoder. This highlights the advantage of multitask learning, where computational resources can be efficiently shared to learn a more effective shared representation. This work also aims to handle data imbalance, which is commonly observed in large-scale datasets. In addition, task balancing for the multitask learning method and computing balanced loss function for both classification and regression are other challenges addressed by this research, thus producing optimal model convergence and better accuracy compared with learning each task individually. Experimental studies show that the proposed model can achieve excellent classification results on the given datasets. The main contribution of this article is summarized as follows.

1) We developed a multitask deep learning model capable of simultaneously predicting various classes and computing continuous values for large-scale datasets.
2) We designed a framework to analyze the behavior of multitask approaches when applied to hyperspectral images characterized by a large number of bands. In addition, this framework evaluates the impact of integrating both spectral and spatial information on the performance of the multitask model.
3) We introduce cost-sensitive learning and focal loss methods for handling class imbalance in data and achieving task balancing for computing multitask loss and optimizing its parameters within a multitask learning framework, aiming to enhance model performance and efficiency for different tasks.

The rest of this article is organized as follows. Section II provides the background and motivation of the proposed work. Section III explains the dataset used and the proposed model of this research work. Section IV presents the quantitative and qualitative results of the proposed model. Furthermore, Section V concludes the research work with future directions.

II. Related Work

In recent decades, many HSI classification and regression frameworks have been proposed for individual tasks due to their broad applications. Traditional methods, such as principal component analysis (PCA) [25], linear discriminant analysis (LDA) [26], local linear embedding [27], sparse representation [28], and manifold learning [29], have been used as feature extraction and dimensionality reduction methods. These methods were proposed to minimize noise and redundancy within the data. However, they require manual parameter tuning by experts and the selection of spectral bands with optimal information content to prevent substantial loss of information. Furthermore, these methods are limited to extracting low-level handcrafted features, making them inadequate for handling unknown or complex scenes [30]. In addition, challenges such as sensor configurations, atmospheric interference, and spectral variations further complicate HSI data analysis, leading to increased intraclass variability and interclass similarity [14], [31]. Consequently, there is a growing demand for more advanced techniques in HSI classification. Recently, learning-based algorithms, particularly deep learning models, have gained significant attention due to their adaptive nature and significant classification performance.

Deep learning models offer the capability to extract high-level abstract features through end-to-end hierarchical frameworks, effectively integrating spectral and spatial information from the input data. Among many deep

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learning algorithms such as extended morphological profiles (EMPs) [32], stacked autoencoder (SAE) [33], deep belief networks (DBNs) [34], recurrent neural networks (RNNs) [35], and generative adversarial networks (GANs) [36], CNNs has shown efficacy in achieving satisfactory performance for HSI classification. The evolution of CNNs has progressed from 1-D CNNs [37], which extract spectral information, to 2-D CNNs [38], incorporating spatial aspects, and eventually to 3-D CNNs [39], which simultaneously consider both spectral and spatial information. Furthermore, to selectively focus on the most important spectral–spatial features among different channels, an attention network is introduced, which has recently been used in many research, specifically in HSI [40], [41], [42]. Although CNNs are generally used for pixel-wise image classification tasks to label different data classes at the image or pixel level, they may not be as effective in predicting continuous variables. In such scenarios, regression-based methods often outperform them [43].

Many researchers have introduced regression-based methods that aim to estimate the quantitative properties of the scene captured by the hyperspectral data. Some common regression techniques applied to hyperspectral data include partial least-squares regression (PLSR) [44], support vector regression (SVR) [45], Gaussian process regression (GPR) [46], and neural network regression (NNR) [47]. These methods are used for tasks such as estimating physical parameters (e.g., vegetation biomass and soil moisture content), predicting environmental variables (e.g., temperature and humidity), and hyperspectral image reconstruction. Khodadadzadeh et al. [48] introduced a multinomial logistic regression method for hyperspectral classification. The accuracy was further improved by using Kronecker factorization-based multinomial logistic regression algorithm proposed by Wang [49]. However, all these different classification and regression methods have been performed as a single-task method. Using a deep learning approach usually yields more parameters, training data, as well as memory and computational costs to generate significant results. Thus, the computational expenses will increase manyfold if we have to deal with multiple tasks one at a time, each demanding significant computational resources and extensive training.

Caruana [50] laid the foundation for applying the multitask learning approach, where he demonstrated that sharing knowledge across related tasks can improve model performance. Multitask learning was initially introduced in the computer vision domain to mitigate computational challenges by concurrently handling multiple tasks through shared representation techniques, thereby enhancing efficiency and reducing resource demands [51], [52]. In deep learning, multitask networks are commonly categorized into hard or soft parameter sharing of hidden layers. In the soft-sharing approach, task-specific parameters are assigned to each task, aiding cross-task interaction for feature sharing [53]. Conversely, the hard-sharing method segregates the network into shared and task-specific parameters [54]. Recently, many methods have been introduced to explore multitask learning in remote sensing applications. Wang et al. [21] proposed a novel boundary-aware multitask learning where boundary detection is employed as an auxiliary task to regularize the other two tasks of semantic segmentation and height estimation. In addition, Guo et al. [55] proposed a scene-driven multitask parallel attention convolutional network (MTPA-Net) to fully utilize building extraction from remote sensing imagery. Improving this method, Hong et al. [56] introduced a multitask learning network for building extraction and change detection from remote sensing images. This method uses a Swin Transformer as a shared backbone network and multiple heads to predict building labels and changes. Liu et al. [23] proposed a hybrid multitask learning framework (HyMuT) by sharing representations across multiple tasks, which is based on the similarity between the data and the target classification task. Most of these methods aim to improve semantic segmentation performance by combining tasks such as edge detection, building height estimation, change detection, and boundary detection. Different from previous remote sensing multitask approaches, our proposed model achieves both classification and regression tasks within a unified framework and further utilizes an automatic task balancing method to compute the multitask loss and to improve their overall performance while reducing the computational costs involved by manual or grid search of the best task weights.

III. Methodology

The encoder–decoder-based U-Net model [57] and its variant have been successfully developed for pixel-wise image classification and regression tasks in remote sensing. Inspired by these models, we propose an end-to-end multitask deep learning model in this study. As shown in Fig. 1, the proposed method consists of three parts: a shared encoder network, a spectral–spatial attention block, and a task-specific decoder. We considered the encoder–decoder architecture to design our proposed model because of its ability to solve image-to-image prediction problems. Considering task correlations, we have considered a standard yet efficient encoder and proposed an original decoding strategy relevant to multitask learning purposes. This section will introduce the dataset considered, discuss data preprocessing and model architecture, and provide detailed insights into the multitask loss algorithms employed in our model.

A. Dataset

We consider the recently proposed Artificial Intelligence dataset for Forest Geographical Applications (TAIGA) hyperspectral dataset [24]. TAIGA is an airborne hyperspectral imagery acquired with the AISA Eagle II hyperspectral line scanner in the southern boreal zone of Finland. It has a spectral resolution of 4.7-nm full-width at half-maximum (FWHM) and a spatial resolution of 0.7 m. It consists of 13 forest variables, among which three are categorical (classification part) and the remaining ten belong to continuous data (regression part), as shown in Table I. It is designed for realistic multitask-learning purposes, having continuous and categorical forest variables. Also, it is large enough to validate models at different scales and reliably identify their generalization behavior.

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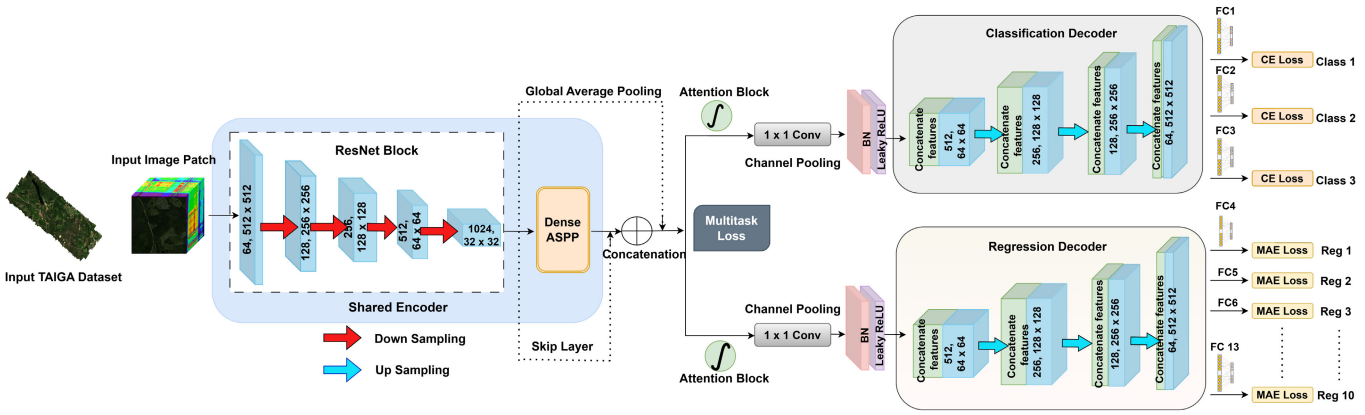


Fig. 1. Framework of the proposed multitask deep learning model for classification and regression of hyperspectral image. Each output has its own loss criterion, CE for classification, and MAE for regression.

TABLE I
FOREST VARIABLES IN THE TAIGA DATASET

S.No.	Variable name	No. of classes or range
	Categorical	
1	Fertility class	4
2	Soil class	2
3	Main tree species	3
	Continuous	
4	Basal area $[m^2/ha]$	$0 - 35.51$
5	Mean DBH [cm]	$0 - 30.89$
6	Stem density [1/ha]	$0 - 6240$
7	Mean height [m]	$0 - 24.16$
8	Percentage of pine	$0\% - 100\%$
9	Percentage of spruce	$0\% - 84\%$
10	Percentage of birch	$0\% - 58\%$
11	Woody biomass [tn/ha]	$0 - 180$
12	Leaf area index	$0 - 9.66$
13	Effective leaf area index	$0 - 6.45$

B. Data Preprocessing

We computed the data distribution for discrete (categorical) and continuous variables in the preprocessing section. Computing the data distribution of different classes in a large dataset is essential for understanding the inherent patterns and imbalances within the data. This information is crucial for developing effective deep learning models, as it helps to address the issues related to class imbalance and optimize training strategies. In addition, analyzing data distribution helps in identifying potential challenges associated with imbalanced datasets, such as the presence of outliers or skewed features. This helps in designing effective data preprocessing strategies and balancing out the majority and minority classes, thus enhancing the model's ability to predict minority classes accurately. In accordance with the original TAIGA dataset article in [24], we excluded categorical and continuous classes that represent less than 0.1% of the total dataset, such as the development class and diameter at breast height. This helps in removing extremely skewed classes while still preserving the overall imbalanced nature of the dataset. Fig. 2 shows the distribution of classes for the categorical variable after removing all the extremely skewed classes.

Furthermore, the regression values of the continuous variables are normalized between 0 and 1 using min–max scaling since the range of values is different for each variable. This transformation brings all the continuous variables to a common scale, ensuring that they share a consistent range. This approach balances the impact of variables with different scales and enhances the interpretability of regression coefficients. Fig. 3 shows the labels of all ten continuous variables.

In addition, we also performed a separability analysis to understand the degree of distinction between classes. This analysis aimed to identify features that play a significant role in separating different classes from one another. We computed region-of-interest (ROI) separability using the Jeffries–Matusita (JM) distance and transformed divergence (TD) method (see Fig. 4). The JM distance quantifies the separability by transforming the multivariate data within each ROI, providing a measure of how well defined the probability distributions of features are between the regions. Higher JM distances indicate increased separability, aiding in effective region classification. On the other hand, the TD method enhances ROI separability by transforming the original data to a space where differences between regions are prominent. This method measures dissimilarity between the probability distributions of features in different ROIs, contributing to a more discriminative analysis.

The main idea is to know the correlation between different variables (continuous and categorical) since all the variables belong to the forest biome and thus have similar spatial resolution. Understanding the degree of distinction between classes is crucial for feature selection and model building to identify the most influential factors in the separation of distinct categories within the dataset. Subsequently, the dataset was split into training, validation, and test sets such that no two samples from the training and test data should overlap each other (see Fig. 5). To reduce spatial autocorrelation and bias in error estimation, the training, validation, and test samples are designed to be spatially disjoint. We adopt the data-splitting approach proposed in [24], which ensures that no spatial overlap exists between these splits. Specifically, the splits are separated by "no label buffers," which are wide enough to guarantee that any input patch centered on a pixel from one

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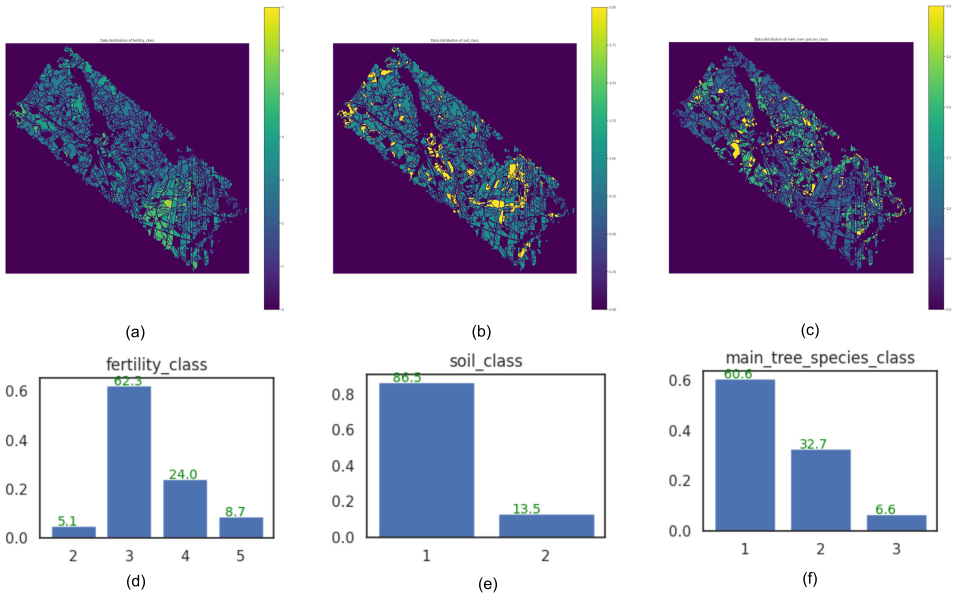


Fig. 2. Visualization of (a)–(c) classification labels and (d)–(f) class distribution for different categorical variables, namely, fertility class, soil type, and main tree species.

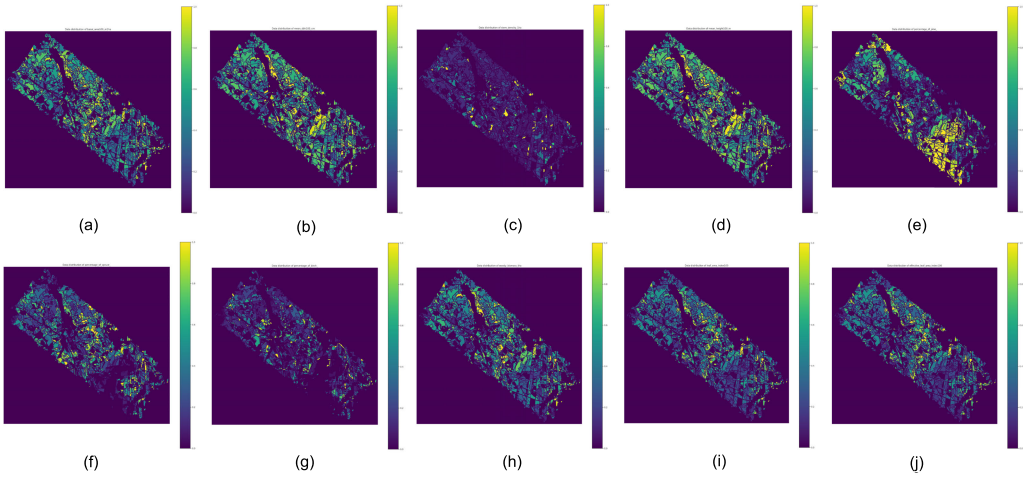


Fig. 3. Visualization of normalized regression labels from 0 to 1 for all ten continuous variables. (a) Basal area, (b) mean dbh, (c) stem density, (d) mean height, (e) % of pine, (f) % of spruce, (g) % of birch, (h) woody biomass, (i) leaf area index, and (j) effective leaf area index.

split (e.g., training) does not extend into neighboring splits (e.g., testing or validation), considering the model's field of view. The input patch size of 45 × 45 pixels is chosen to capture essential spatial details while maintaining the integrity of the splits. Furthermore, Mottus et al. [58] and Cu [59] conducted a detailed analysis with the TAIGA dataset, strictly forbidding any spatial overlap between training and test sets, ensuring unbiased and reliable error estimation. Our work replicates this methodology for consistency and comparability.

C. Shared Encoder Block

An encoder is a crucial component in neural network architectures, transforming input data into a compressed latent representation and capturing essential features while discarding irrelevant details with respect to the objectives. The proposed shared encoder consists of a residual network (ResNet) block and a dense atrous spatial pyramid pooling (DenseASPP), which are well-established and widely used building blocks in deep learning architectures.

1) Residual Network: ResNets [60] is a type of deep neural network architecture that addresses the vanishing gradient problem by introducing shortcut connections, allowing the model to learn residual mappings. These shortcuts facilitate the training of very deep networks, improving performance in tasks such as image classification and regression tasks. The choice of the ResNet model for the shared encoder in our architecture is driven by its intrinsic advantages in training deep neural networks and achieving state-of-the-art performance across various tasks, including multitask learning models. ResNet consists of residual blocks that enable the training of networks with hundreds of layers, which is critical for tasks requiring complex feature extraction and classification. Compared to alternative architectures such as Visual Geometry Group (VGG), Inception, DenseNet, or MobileNet, ResNet's combination of depth, performance, transfer learning capabilities, and extensive empirical validation makes it an optimal choice for our architecture's shared encoder. The availability of resources, tools, and optimizations further

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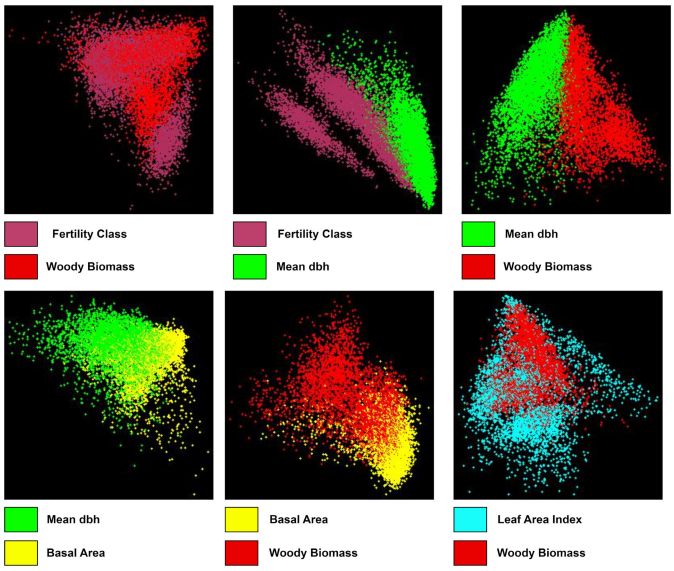


Fig. 4. Visualization of ROI separability to know the class overlap and correlation between different forest variables (continuous and categorical).

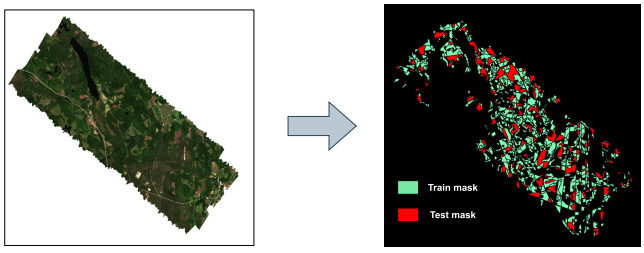


Fig. 5. Split dataset into training and test sets such that no two samples overlap each other and not a single pixel should present in both training and testing sets.

supports our decision to utilize ResNet to enhance model efficacy and efficiency.

As shown in Fig. 6, the encoder network begins by employing a convolution block to augment the number of feature dimensions. This block comprises a 2-D convolution layer with a 3×3 kernel size, followed by a batch normalization layer and an activation layer. The choice of a smaller kernel size aims to prevent information loss from input images by summarizing features across pixels. Batch normalization is applied to stabilize and expedite training, while the activation layer introduces nonlinearity to the model. Leaky ReLU is selected as the activation function instead of the commonly used ReLU. This decision is motivated by the observation that neurons cease learning when the ReLU function enters the negative range. Leaky ReLU addresses this limitation by allowing a small negative slope for negative inputs.

Following this is the cascade of the ResNet. Let $R(x)$ represent the final output of the ResNet; then,

$$F(x) = R(x) - x \tag{1}$$

where x denotes the input image and $F(x)$ represents the residual mapping (the dotted line in Fig. 6). The abovementioned equation is called a shortcut connection in

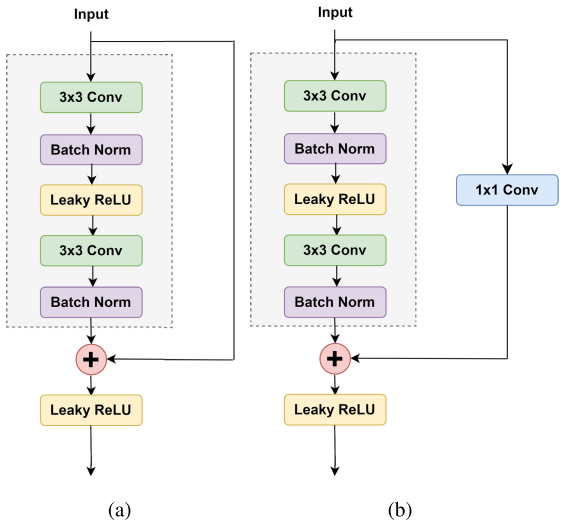


Fig. 6. ResNet block. (a) Identity ResNet block and (b) ResNet block with a 1×1 convolution that transforms the input into the desired shape for the addition operation called "bottleneck block."

a feedforward network and can also be represented as

$$y_i = F(x_i, \{w_i\}) + x_i \tag{2}$$

where y_i and x_i represent the output and input vectors of the ith set of stacked layers, respectively, and $F(x_i, \{w_i\})$ is the shortcut connection. For consistency with the input tensor x, the shortcuts in our model must have equal sizes. These shortcuts can be mapped as an identity or through a linear projection to match dimensions. Fig. 6(b) showcases a ResNet block, a key component in our network construction. Within the ResNet block, each convolutional layer is followed by a batch normalization. Our proposed model adjusts the number of channels in x_i within the ResNet block. Consequently, our model employs the ResNet block with a projection shortcut with 1×1 conv layer as it can learn more complex features specifically on larger datasets and have better scalability.

2) Dense ASPP: A multiscale approach is required since the accurate analysis of complex tasks, such as simultaneous classification and regression, demands robust feature representation across various spatial resolutions. In our proposed framework, the integration of Dense ASPP [61] is instrumental in achieving this goal effectively. DenseASPP is an extension of the atrous spatial pyramid pooling (ASPP) module, which captures multiscale contextual information. ASPP employs atrous (dilated) convolutions at multiple dilation rates, thereby enabling the network to perceive both local and global context information simultaneously. Initially developed for semantic segmentation, ASPP has proven its versatility in various network applications by efficiently extracting multiscale features to attain a substantial receptive field. In our study, we advanced the ASPP module by incorporating dense connectivity, resulting in the Dense ASPP modification. Its dense connectivity enhances feature reuse across tasks, leading to improved model efficiency and performance in jointly learning classification and regression tasks, all achieved without a significant increase in computational cost.

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The atrous convolution can be formulated as follows:

$$o[i] = \sum_{k=1}^{F} y[i + r.k].w[k] \tag{3}$$

where o and y are the output and input signals (output of ResNet), respectively; F is the filter size; $w[k]$ represents the kth parameter of the filter; and r is the dilation rate. Each atrous layer's result is combined with the input feature map and the outputs from preceding lower layers. This concatenated feature map is then forwarded to the subsequent layer in the network. The ultimate output of Dense ASPP comprises a feature map produced through the utilization of multirate, multiscale atrous convolutions. This innovative architecture enables the simultaneous construction of a more comprehensive and intricate feature pyramid with minimal dependence on the extensive use of atrous convolutional layers. The Dense ASPP module is typically expressed as a concatenation of dilated convolutions with different dilation rates. The mathematical expression for a Dense ASPP module with dilation rates $r_1, r_2, \ldots, r_n$ can be written as

$$Y = \text{Concat}\big(\text{Conv}_{r_1}\big(X, W_{r_1}\big), \ldots, \text{Conv}_{r_n}\big(X, W_{r_n}\big)\big) \tag{4}$$

where $\text{Conv}_{r_1}(X, W_{r_1})$ denotes the dilated convolution with dilation rate r_n applied to the input feature map X using the corresponding set of filters W_{r_i}. The Concat$(\cdots)$ operation combines the feature maps obtained from these dilated convolutions along the channel dimension. The dilation rate considered for the proposed work are 6, 12, and 18 (see Fig. 7).

DenseASPP, through its integration of ASPP and dense connectivity, aims to improve the performance of the network by integrating contextual information, making it suitable for tasks like classification and regression. The ASPP module enhances the model's ability to capture contextual information, proving valuable for classification and regression by considering broader understanding. Meanwhile, dense connectivity ensures efficient training by addressing issues like the vanishing gradient problem. Furthermore, the flexibility in architecture design meets the specific requirements of different tasks within a multitask learning framework. We adopt summation instead of concatenation because summation reduces feature dimensions and the number of parameters in the next layers, thus simplifying model and making training easier. The integration of multiple reception information from ResNet and Dense ASPP can be formulated as follows:

$$F_l = f_{l-1} + \sum_{i=1}^{N} \frac{1}{1 + e^{-\lambda_i}} \times \text{BN}\Big(w^{c}_{1\times1,0}\Big(A\Big(\text{BN}\Big(w^{1.5.h}_{3\times3,d_i}(f_{l-1})\Big)\Big)\Big)\Big) \tag{5}$$

where f_{l-1} denotes the input features and c denotes the number of bands of the input features. Batch normalization (BN) $(\cdot)$ and $A(\cdot)$ represent the batch normalization operation and Leaky ReLU activation function, respectively. $w^{1.5.h}_{3\times3,d_i}$ denotes 3×3 convolutional kernels with d_i as the dilation rate and $1.5.h$ as the number of bands of the output features. $w^{c}_{1\times1,0}$ indicates 1×1 convolutional kernels with no dilation, and the number of bands of the output features is equal to h. The output of the summation layer is subjected to a global average-pooling layer to halve the original size in the spatial dimensions, followed by a skip layer to increase the feature channels to twice the amount of the input channels, which can be formulated as follows:

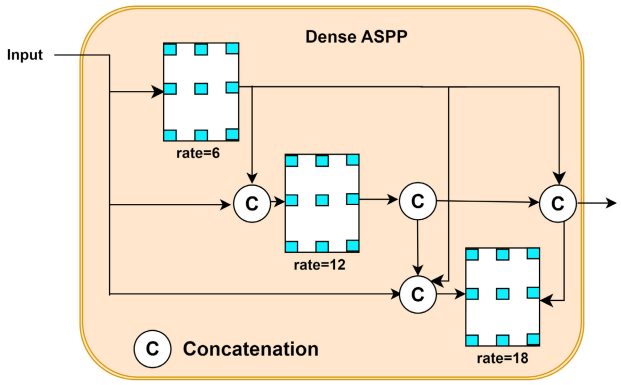


Fig. 7. Architecture of Dense ASPP with dilation rate of 6, 12, and 18.

$$f_l = A\big(\text{BN}\big(w^{2\cdot c}_{1\times1,0}(\text{GAP}(F_l))\big)\big) \tag{6}$$

where global average pooling (GAP) $(\cdot)$ and f_l denote the average-pooling layer and the output of the summation, respectively. $w^{2\cdot c}_{1\times1,0}$ denotes 1×1 convolutional kernels with no dilation, and the number of bands of the output features equals $2 \cdot c$.

D. Spectral–Spatial Residual Attention Block

The attention network in deep learning models is introduced to focus more on important features and discard unnecessary features [62]. By enabling selective information processing, these modules allow the model to dynamically focus on relevant parts of the input for each task, enhancing its adaptability to different requirements. The ability to selectively attend to task-specific features improves generalization, making the model more robust to unseen data and tasks. In a typical attention network, a weight matrix is generated from the input and is applied back to itself. Woo et al. [63] introduced a convolutional block attention module (CBAM), a simple yet effective attention network for CNNs. An attention network can be formulated as follows:

$$R_l = R_{l-1} \cdot A(R_{l-1}) \tag{7}$$

where R_l and R_{l-1} represent the input and output of the residual unit of lth layer, respectively, and the function $A(\cdot)$ represents operations such as pooling, fully connected, ReLU, and sigmoid function. In this work, we have proposed the combination of attention and ResNet called residual attention network to enhance feature representation and improve gradient flow, which can be represented as

$$R_l = R_{l-1} + F(R_{l-1}) \cdot A(F(R_{l-1})) \tag{8}$$

where F represents the feature map. As shown in Fig. 8, the sequential feeding of input features into spectral attention modules and spatial attention modules is a strategic approach aimed at empowering learning-based networks to discover both "what" and "where" to focus during information processing. A channel attention map is generated by first

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passing the input through a spectral attention module, allowing the network to grasp "what" to prioritize in the channel dimension. This mechanism enhances the model's ability to selectively emphasize relevant features within each band, providing a foundation for capturing critical patterns and information. The subsequent utilization of spatial attention modules further refines this process, enabling the network to understand "where" in the spatial domain to concentrate its attention.

Within the channel attention module, the input features undergo a dual transformation through a max-pooling layer and an average-pooling layer, resulting in two distinct vectors. These pooling feature maps are combined and fed as input to the attention network and are calculated as follows:

$$Z = P_{\text{avg}} + P_{\max} \tag{9}$$

where

$$P_{\text{avg}} = \text{Avg}(F) = \frac{1}{h \times w} \sum_{i=1}^{h} \sum_{j=1}^{w} F_{i,j} \tag{10}$$

$$P_{\max} = \max(F). \tag{11}$$

A feature map $F \in \mathbb{R}^{h \times w \times d}$, where $h \times w$ denotes the spatial size, d denotes number of spectral bands, and $F_{i,j}$ denotes the value at position (i, j) of the input F. A 1-D CNN is used for mapping spectral vectors and extracting spectral features, followed by a shared multilayer perception (MLP). Two bottleneck fully connected MLP layers are considered to enhance the extracted spectral features. This layer tends to reduce model complexity and support model generalization ability. The first layer, represented as W_1, is a dimensionality reduction layer, and the second layer, represented as W_2, is a dimensionality increment layer. The spectral features are calculated and can be formulated as follows:

$$F_{\text{spectral}} = \sigma\left(W_2\left(W_1\left(P_{\text{Avg}}\right)\right)\right) + W_2(\text{ReLU}(W_1(P_{\text{Max}}))) \tag{12}$$

where σ represents the sigmoid function and ReLU is the hidden activation of the shared MLP. Spatial attention maps are computed using max pooling and average pooling along the channel axis to generate two feature maps. These feature maps are concatenated and processed through a 3×3 convolution layer, followed by a sigmoid function to limit attention values between 0 and 1. The final step involves a multiplication operation between the input features and the 2-D spatial attention map along the spatial dimensions. The rich low-/mid-level features are transmitted to the decoder portion through attention modules in the channel pooling layer. This is performed before the first upsampling step, where upscaled features are compressed separately for regression and classification tasks to 256 dimensions with a 1×1 kernel. As shown in Section III-B, classification and regression tasks indeed need decorrelation that facilitates the amplification of meaningful information while suppressing irrelevant features. The channel pooling layer captures complex patterns or relationships between different channels in the feature map by adjusting the weights in the 1×1 convolutions with adaptive channel interactions.

E. Task-Specific Decoder

In the multitask learning model, task-specific decoders allow the individual adjustment of parameters, accommodating variations in task complexity and requirements. This capability to independently customize adjustments aids in isolating task-specific features throughout the training process, thereby effectively reducing interference between tasks. We apply batch normalization to each layer to concatenate the resulting features together. This produces the shared representation between each task. A Leaky ReLU activation function follows this to improve the nonlinear learning ability of the model. Once all shared features are extracted, the multitask decoder is designed, and we first separate the classification and regression tasks into two distinct paths, each being more task-specific.

We deliberately kept every decoder block symmetrical to the shared encoder block, and the features of the corresponding encoding layer concatenated with the task-specific decoder (see Fig. 1). This is to maintain the multiscale features, enhance important channel features, and weaken unimportant channel features. As mentioned in Section III-B, computing class overlap and feature correlation helps in designing dedicated ResNet-based decoders for each path by identifying and utilizing common features that are beneficial for both classification and regression tasks. This approach allows each decoder to specialize in refining these shared features according to the specific needs of each task by reducing task interference. Finally, task-specific heads are optimized, relying on a single fully connected layer. Bilinear upsampling is applied to scale the output to the same resolution as the input. Most of the model's parameters and depth are thus in the feature encoding, with very little complexity in each task decoder. This highlights the advantage of multitask learning, where computational resources can be efficiently shared to learn a more effective shared representation. By employing task-specific loss functions, such as cross entropy (CE) for classification and mean absolute error (MAE) for regression, the model optimizes parameters independently for each task, enhancing training efficiency.

F. Multitask Loss

Loss functions are crucial in determining how models assess the overall discrepancy between predicted outcomes and ground truth. The selection of a loss function has a direct impact on both the learning process and the ultimate outcomes of models. Even a well-designed model can be affected if paired with an inappropriate loss function. Furthermore, when considering the multitask model, loss balancing must be studied. The multitask loss function combines the individual losses associated with each task into a single composite loss, which the model aims to minimize during training. The formulation of a multitask loss typically involves assigning weights to each task's contribution to the overall loss, reflecting the relative importance of different tasks. This allows the model to strike a balance between optimizing for various multitask objectives such as classification and regression. In this research, the multitask loss function has been used for different purposes: 1) to handle class imbalance and 2) task balancing for multitask learning.

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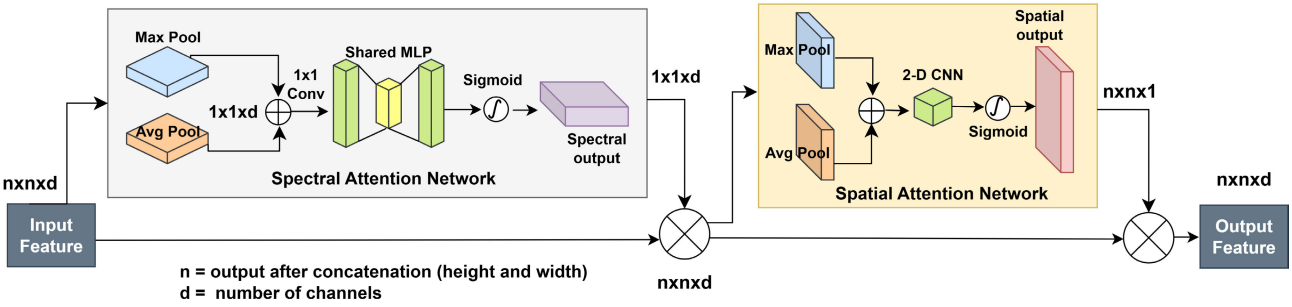


Fig. 8. Framework of spectral–spatial residual attention block based on CBAM.

In the given dataset, the class distribution is not uniform, i.e., there are a few minor classes that are less than 10% of the whole dataset. This leads to data imbalance, resulting in the consideration of major classes and the ignoring of minor classes. In dealing with class imbalance, it is crucial to appropriately adjust learning goals and metrics. Attempts to address class imbalance often involve sampling strategies, such as oversampling minority classes or undersampling majority classes. However, these approaches have limitations. Oversampling can lead to overfitting, while undersampling may discard valuable information. In multilabel, multiclass scenarios, the complexity increases as a sample may belong to a minority class for one label and a majority class for another, making traditional resampling less effective. Thus, in this work, to handle the class imbalance in the dataset, instead of using a sampling method, two different methods have been adopted, naming cost-sensitive learning [64] and focal loss [65] (without concurrent methods).

Cost-sensitive learning considers the misclassification cost, thus reducing the performance accuracy in imbalanced datasets where the algorithm is biased toward majority classes. Let c_i be the cost of classifying a sample for class i and p_i be the frequency of class i in the dataset. The cost for the wrong prediction for class i is computed as [66]

$$c_i = \frac{1}{p_i}. \tag{13}$$

However, this does not work for an extremely imbalanced dataset. For our multitask imbalanced dataset, we considered inverse median frequency for computing classification cost. If p_m is the median of class frequencies, then the cost for the wrong prediction for class i is computed as

$$c_i = \frac{p_m}{p_i}. \tag{14}$$

The standard CE loss assumes that all classification errors are equally important, which fails for class-imbalanced datasets. Thus, the focal loss dynamically scales the standard CE loss. The focal loss [65] for a multitask imbalanced dataset can be computed as

$$\iota_f = -\alpha(1-p)^{\gamma}\log p \tag{15}$$

where α is the balanced factor and γ is the focusing parameter. With reference to the work in [67], we considered the value of α as 0.25 and γ as 2. $p \in [0, 1]$ is the predicted value of the given data sample.

In addition, training the multitask model requires optimizing each loss function weight to improve the overall performance accuracy. We have considered two different approaches to address this issue: uncertainty loss and gradient normalization (GradNorm) loss.

1) Uncertainty Loss: The uncertainty loss of every single task in classification and regression can be computed using Cipolla et al. [54] assumption for the multitask loss function. This method, known as Homoscedastic uncertainty loss, refers to a type of uncertainty that remains constant regardless of input data. Instead, it varies between tasks and is task-dependent. In multitask learning, homoscedastic uncertainty can effectively capture the relationship between related tasks and thus aim to fine-tune task weightings through probabilistic modeling.

Consider $f(x; W)$ as the output of the deep neural network, where W is the weight and x is the input. Let the Gaussian distribution be the considered likelihood for the regression model with mean $f(x; W)$ and variance σ^2 given as

$$p(y|f(x; W)) = N\big(f(x; W), \sigma^2\big). \tag{16}$$

The output likelihood is typically passed through a softmax function for classification tasks where Cipolla et al. [54] utilized a scaled version of the model output given as

$$p(y|f(x; W), \sigma) = \text{Softmax}\left(\frac{1}{\sigma^2} f(x; W)\right) \tag{17}$$

where σ is a scalar noise. The model is optimized by maximizing the log likelihood of its output. For regression tasks, the log likelihood of the output is expressed as follows:

$$\log p(y|f(x; W)) \propto -\frac{1}{2\sigma^2}\|y - f(x; W)\|^2 - \log\sigma \tag{18}$$

and for classification, the output is

$$\begin{aligned}\log p(y|f(x; W), \sigma) &= \log \text{Softmax}\left(\frac{1}{\sigma^2} f(x; W)\right) \\ &= \log \frac{\exp\left(\frac{1}{\sigma^2} f_c(x; W)\right)}{\sum_{c'} \exp\left(\frac{1}{\sigma^2} f_{c'}(x; W)\right)} \\ &= \frac{1}{\sigma^2} f_c(x; W) \\ &\quad - \log \sum_{c'} \exp\left(\frac{1}{\sigma^2} f_{c'}(x; W)\right)\end{aligned} \tag{19}$$

where $f_c(x; W)$ is the cth element of vector $f(x; W)$

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Let the output for regression be y_1 and for classification be y_2. The multitask loss can be given jointly as

$$\begin{aligned}
\pounds(W,\sigma_1,\sigma_2) &= -\log p(y_1, y_2 = c|f(x;W)) \\
&= -\log N(y_1; f(x;W), \sigma_1^2) \\
&\quad -\log p(y_2 = c|f(x;W),\sigma_2) \\
&= \frac{1}{2\sigma_1^2}\|y_1 - f_c(x;W)\|^2 + \log\sigma_1 \\
&\quad -\frac{1}{\sigma_2^2} f_c(x;W) \\
&\quad +\log\sum_{c'}\exp\left(\frac{1}{\sigma_2^2} f_{c'}(x;W)\right) \\
&= \frac{1}{2\sigma_1^2}\pounds_1(W) + \log\sigma_1 - \frac{1}{\sigma_2^2} f_c(x;W) \\
&\quad +\log\sum_{c'}\exp\left(f_{c'}(x;W)\right)^{\frac{1}{\sigma_2^2}} \\
&\quad +\log\sum_{c'}\exp\left(\frac{1}{\sigma_2^2} f_{c'}(x;W)\right) \\
&\quad -\log\sum_{c'}\exp(f_{c'}(x;W))^{\frac{1}{\sigma_2^2}} \\
&\approx \frac{1}{2\sigma_1^2}\pounds_1(W) + \log\sigma_1 + \frac{1}{\sigma_2^2}\pounds_2(W) + \log\sigma_2
\end{aligned} \tag{20}$$

where

$$\pounds_1(W) = \|y_1 - f_c(x;W)\|^2$$

and

$$\pounds_2(W) = -\log \text{Softmax}(y_2, f_c(x;W)).$$

The final equation can be derived assuming

$$\frac{1}{\sigma_2^2}\sum_{c'}\exp\left(\frac{1}{\sigma_2^2} f_{c'}(x;W)\right) \approx \sum_{c'}\exp(f_{c'}(x;W))^{\frac{1}{\sigma_2^2}}$$

when $\sigma_2 \to 1$.

The objective function shows that σ_1 and σ_2 control how much influence $\pounds_1(W)$ and $\pounds_2(W)$ have on the overall loss. This approach can be straightforwardly extended to handle multiple regression and classification outputs. In terms of implementation, the numerical stability is managed by training the logarithm of the variance given as

$$s = \log\left(\sigma^2\right) \tag{21}$$

where s is the uncertainty loss and σ is the log variance for likelihood of the output.

2) GradNorm Loss: It is another loss-based regularization function that handles task balancing in the multitask learning model. It automatically balances the multitask loss function by directly tuning the gradients to equalize task training rates. The GradNorm is computed as [68]

$$\iota_{\text{grad}}(t; w_i(t)) = \sum_i \left\| G_w^i(t) - \bar{G}w(t).[r_i(t)]^\alpha \right\| \tag{22}$$

where $G_w^i(t)$ is the $L2$ norm of the gradient for a task i of a weighted task loss w at time step t, $\bar{G}w(t)$ is the average gradient norm, $r_i(t)$ is a relative inverse training rate, and α is a hyperparameter that modulates the strength of training rate balancing. The pseudocode of the GradNorm algorithm is shown in Algorithm 1. GradNorm proved to be an efficient algorithm for multitask learning.

Algorithm 1 Training With GradNorm

```
Initialize w_i(0) = 1∀i
Initialize network weights 𝒲
Pick value for α > 0 and pick the weights W (usually the
    final layer of weights which are shared between tasks)
for t = 0 to max_train_steps do
    Input batch x_i to compute L_i(t)∀i and
        L(t) = Σ_i w_i(t)L_i(t) [standard forward pass]
    Compute G_W^(i)(t) and r_i(t)∀i
    Compute Ḡ_W(t) by averaging the G_W^(i)(t)
    Compute L_grad = Σ_i | G_W^(i)(t) − Ḡ_W(t) × [r_i(t)]^α |_1
    Compute GradNorm gradients ∇_{w_i} L_grad , keeping
        targets Ḡ_W(t) × [r_i(t)]^α constant
    Compute standard gradients ∇_𝒲 L(t)
    Update w_i(t) ↦ w_i(t + 1) using ∇_{w_i} L_grad
    Update 𝒲(t) ↦ 𝒲(t + 1) using ∇_𝒲 L(t) [standard
        backward pass]
    Renormalize w_i(t + 1) so that Σ_i w_i(t + 1) = T
end for
```

IV. Experiments and Result Analysis

A. Experimental Configuration and Parameter Settings

In this article, the experiment has been conducted on a computer with an Intel[1] Xeon[1] Silver 4214R CPU at 2.40 GHz with 64-GB RAM and an NVIDIA GeForce RTX A6000 graphical processing unit (GPU) with 51-GB RAM. The software environment is the Ubuntu 14.04 ultimate 64-bit system with a deep learning framework of PyTorch. Deep learning relies on data and is prone to overfitting. To avoid overfitting, we implemented data augmentation (random horizontal and vertical flips), ensuring that each batch of images remained diverse. Following augmentation, we standardized the images by normalizing them using min–max scaling for precalculated mean and variance. We generated prediction maps for the test dataset during the inference stage without employing data augmentation. We initialized the learning rate at 0.001 and the Adam optimizer as a stochastic gradient descent method with a momentum of 0.7 and weight decay (η) of 10^{-4}. We employed a learning rate warm-up strategy for the initial five epochs to enhance model stability during training. This involved setting the learning rate in the first epoch to one-fifth of the initial value and incrementing it by 0.0002 per subsequent epoch until reaching a learning rate of 0.001. Full-resolution images have been used to train the network for 150 epochs with a mini-batch of size 64.

In networks designed for multitask learning, establishing an appropriate balance among various classes is crucial when

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training multiple tasks. To assess the effectiveness of our approach, we conducted training using the proposed method with pretrained parameters and a frozen encoder network. We used CE loss for the classification tasks and MAE loss as the loss function for the regression tasks. This is to measure the gap between the ground truth and the prediction map and to update the network parameters through backpropagation. We opted for the CE loss in the context of classification tasks at the pixel level, as it is well suited for scenarios where the goal is to measure the difference between predicted probability distributions and actual class distributions. The CE can be represented as

$$\text{Cross-Entropy} = -\frac{1}{N}\sum_{j=1}^{N}\sum_{i} y_{ji}\cdot\log\left(\hat{y}_{ji}\right) \tag{23}$$

where N is the number of samples, y_{ji} represents the true label, and $\hat{y}_{ji}$ represents the predicted probability for the ith class of the jth data point. On the other hand, the MAE loss was chosen for regression tasks as it measures the average magnitude of errors between predicted values and actual values. In regression problems, the goal is to predict continuous numerical values, and MAE is a suitable choice for quantifying the accuracy of predictions by emphasizing the magnitude of errors. In addition, given that the data are scaled to small values, MAE becomes more effective in addressing small errors. The MAE can be represented as

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{24}$$

where n is the total number of data points, y_i is the true value, and $\hat{y}_i$ is the predicted value for the ith data point.

B. Experimental Results

1) Quantitative Results: In this research, we have employed six standard evaluation metrics for categorical variables (classification) and three standard evaluation metrics for continuous variables (regression) to systematically assess the effectiveness of the proposed method. For categorical variables, we have computed precision, recall, F1-score, area under the receiver operating characteristic curve (ROC-AUC) score, overall accuracy (OA, micro), and mean class accuracy (macro). The confusion matrix between the predicted class and ground truth labels was calculated first, which determined the number of four classifying conditions on pixel level: true class prediction for a given categorical variable on a positive sample (TP), false class prediction for a given categorical variable on a positive sample (FP), true class prediction for a given categorical variable on a negative sample (TN), and false class prediction for a given categorical variable on a negative sample (FN). OA represents the percentage of correctly classified pixels (summation of TP and TN) over the total number of pixels.

Precision: It is the ratio of true positive predictions to the total predicted positives. It measures the accuracy of positive predictions

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives + False Positives}}. \tag{25}$$

Recall: It is the ratio of true positive predictions to the total actual positives. It measures the ability of the model to capture all the positive instances

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives + False Negatives}}. \tag{26}$$

F1-Score: It is the harmonic mean of precision and recall. It provides a balance between precision and recall

$$F1\text{-Score} = \frac{2\times(\text{Precision}\times\text{Recall})}{\text{Precision + Recall}}. \tag{27}$$

ROC-AUC Score: ROC-AUC measures the area under the ROC curve, which is a graphical representation of the tradeoff between true positive rate (sensitivity) and false positive rate (1 − specificity).

OA (Micro-Average): Micro-average accuracy considers the overall performance across all classes by aggregating individual class metrics

$$\text{Micro Accuracy} = \frac{\text{Sum of True Positives across all classes}}{\text{Sum of (True Positives + False Positives)}}. \tag{28}$$

Mean Class Accuracy (Macro-Average): Macro-average accuracy calculates the average accuracy for each class and then averages these values

$$\text{Macro Accuracy} = \frac{\text{Average of Class Accuracies}}{\text{Number of Classes}}. \tag{29}$$

Microaccuracy favors the model's good performance for the majority class or classes, whereas macroaccuracy also gives equal weight to the minority classes. We have computed standard metrics for different categorical classes in Table II. We performed the metric computation for multiple runs (at least ten times with different random seeds initialized) for each class to know the fluctuation of accuracy in our model. Repeatedly computing accuracy for the proposed model across multiple runs and calculating the minimum and maximum values serves several crucial purposes. First, it addresses the randomness inherent in data splitting, particularly when employing techniques like random sampling or shuffling. In addition, it accounts for the variability that certain machine-learning models may exhibit due to stochastic elements or random initialization.

The practice of assessing accuracy multiple times aids in understanding the model's robustness, as extreme values may highlight sensitivity to specific conditions or data subsets. This approach also facilitates outlier detection, revealing potential issues within the dataset. When tuning hyperparameters, analyzing the range of accuracies helps gauge the model's sensitivity to different settings, informing stable configurations. Furthermore, statistical analysis, including metrics like mean class accuracy performed on multiple runs, offers insights into the central tendency and variability of the model's performance. Importantly, this method simulates real-world scenarios where the model may encounter diverse data variations, providing a more realistic assessment of its performance in different situations. From Table II, it can be observed that the model's performance is consistent, which implies that the model's behavior

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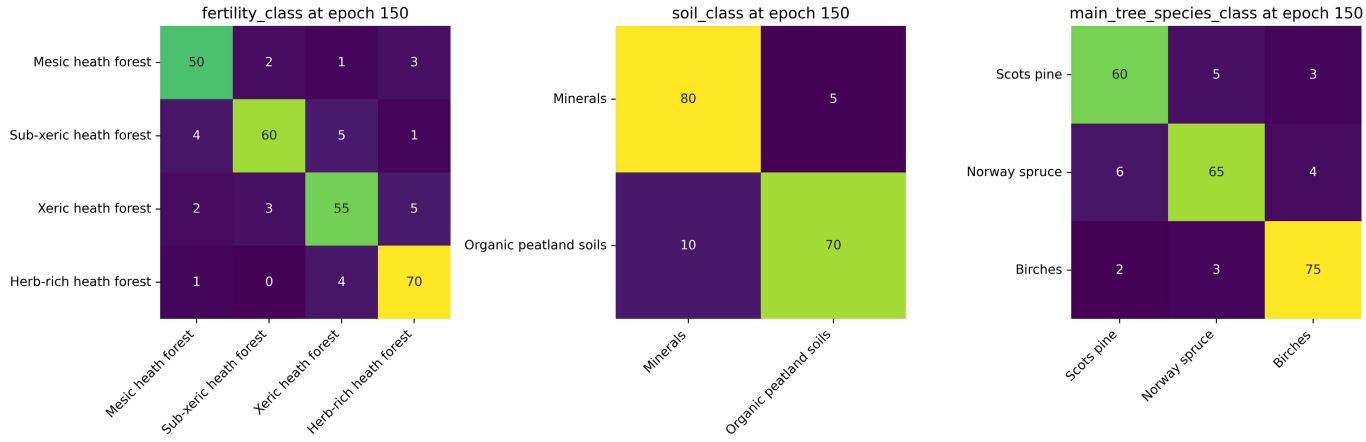


Fig. 9. Normalized confusion matrices for the classification tasks. The horizontal axis shows the true classes, and the vertical axis shows the predicted classes.

is reproducible. It also indicates good generalization to unseen data. Furthermore, we computed normalized confusion matrices for the classification tasks at the end of the training process, as shown in Fig. 9. The model was evaluated on the validation set every ten epochs to monitor its learning progress. Initially, the model tried to predict the minority class due to significant repercussions from the classification loss function. However, as the training advanced and the model encountered the training samples repeatedly, its performance significantly improved by the conclusion of the training period.

For continuous class (regression method), we have considered statistical computation such as the root-mean-square error (RMSE), MAE, and coefficient of determination (R^2; see Table III). Among all other regression methods, RMSE is chosen for its emphasis on larger errors, providing a meaningful measure of the average magnitude of deviations in a regression model, particularly when larger errors are critical. MAE is favored for its simplicity and interpretability. It offers a straightforward average of absolute errors without emphasizing extreme values, making it suitable for scenarios where all errors are considered equally important. R^2 is a widely chosen metric for its comprehensive evaluation of model performance. It captures the proportion of variance explained by the model, providing a global measure of fit and predictive power. The equation for these computational methods is given as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{30}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{31}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{32}$$

where n is the total number of data points, y_i is the true value, and $\hat{y}_i$ is the predicted value for the ith data point. $|\cdot|$ represents the absolute value for the MAE in (31). A lower RMSE indicates better precision, with values closer to 0 suggesting a more accurate model. Similarly, smaller MAE values indicate better accuracy and reduced bias in the model. In contrast, a higher R^2 indicates a better fit, with 1.0 indicating perfect prediction and 0.0 indicating a model that does not explain any variance. In Table III, it can be observed that our proposed model demonstrates higher mean performance while maintaining lower or equivalent variability, as evidenced by reduced differences between the minimum and maximum values of RMSE, MAE, and R^2 when trained across ten different seeds. The ten different trials/seeds of the experiment were conducted to ensure that the reported performance metrics are robust and represent the model's true capability.

2) Qualitative Results: For the qualitative evaluation, we have a classification map for the categorical variable to visualize different classes of the TAIGA dataset, as shown in Fig. 10. The classification map visually represents the model's predictions, assigning specific class labels to different regions. This helps in validating the accuracy of the classification models by comparing them with ground truth data for validation and accuracy assessment. A subset image from the full TAIGA dataset has been chosen to capture all the species of different classes and ease visualization and interpretation.

For the regression method, scatter plots and kernel density plots between prediction and target label have been plotted on randomly selected continuous forest variables (see Fig. 11). The scatter plot represents the target labels on the x-axis and the predicted labels on the y-axis. Ideally, all points should align with the line $y = x$ or the black dotted line on the plot. The top marginal plot illustrates the true distribution, while the right marginal plot represents the predicted distribution. In addition, kernel density estimation plots specify the density of points on the respective scatter plots, with darker regions indicating higher point concentrations. A robust model is characterized by darker regions surrounding the modes of the target distribution. The Pearson correlation coefficient is also computed to know how closely the predicted and target outputs are correlated. High Pearson correlation coefficients in the scatter plots ($\geq$0.93) prove that the prediction and target labels for all the regression tasks are strongly correlated. In addition, an error histogram (see Fig. 12) is plotted by computing the absolute error for each test sample and plotting the histogram

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TABLE II

PERFORMANCE EVALUATION FOR ALL THREE DIFFERENT CATEGORICAL VARIABLES. TO ENSURE THAT THE REPORTED PERFORMANCE METRICS ARE ROBUST AND REPRESENTATIVE OF THE MODEL'S TRUE CAPABILITY, WE HAVE CONDUCTED TEN TRIALS/SEEDS OF THE EXPERIMENT TO GET MIN/MAX/MEAN VALUES

Categorical	Precision	Recall	F1-score	ROC-AUC score	Overall Accuracy (Micro)	Mean Class Accuracy (Macro)
Fertility Class	94.66 Max: 94.72 Min: 94.50	94.12 Max: 94.83 Min: 94.03	94.23 Max: 94.79 Min: 94.09	96.89 Max: 96.98 Min: 96.72	97.03 Max: 97.13 Min: 96.96	95.78 Max: 95.93 Min: 95.10
Soil Class	98.57 Max: 98.63 Min: 98.29	98.51 Max: 98.68 Min: 98.18	98.53 Max: 98.59 Min: 98.21	98.25 Max: 98.54 Min: 98.02	98.73 Max: 98.49 Min: 98.13	98.26 Max: 98.76 Min: 98.16
Mean Tree Species	97.36 Max: 97.53 Min: 97.06	97.29 Max: 97.78 Min: 97.08	97.31 Max: 97.49 Min: 97.15	97.89 Max: 97.96 Min: 97.81	97.91 Max: 97.98 Min: 97.64	97.55 Max: 97.84 Min: 96.98

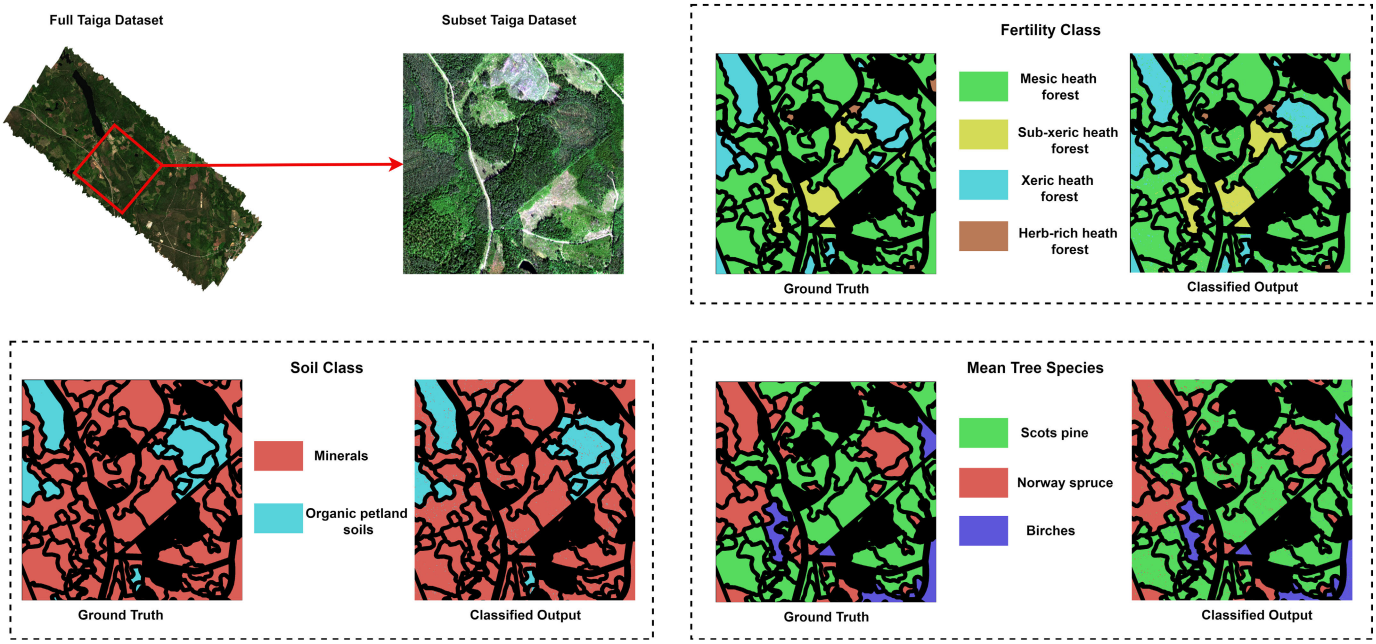
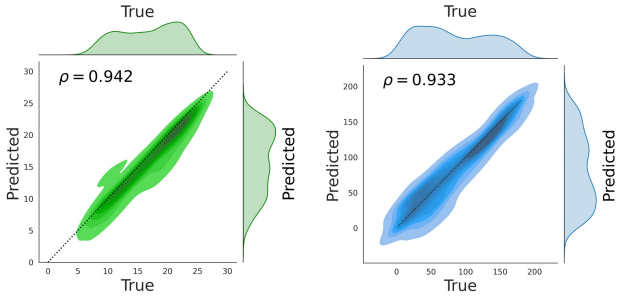


Fig. 10. Pixel-wise classification map for all three categorical variables to visualize different classes of the TAIGA dataset.



(a) (b)

Fig. 11. Prediction accuracy for continuous forest variable. (a) Mean tree height and (b) woody biomass with Pearson correlation coefficient of 0.942 and 0.933, respectively.

of the error values with 100 bins. A good model will have most of the errors falling in bins that are close to zero. The error histograms generated by our models show that the models are in close proximity to zero. This gives an overview of how well the model performs in general.

C. Comparison Analysis

In Table IV, we categorize the results into three sections to comprehensively assess the performance of our proposed model: single-task classification, single-task regression, and multitask learning, covering both classification and regression simultaneously. This division allows us to thoroughly evaluate the model's capabilities in specialized tasks individually and then in a multitask learning approach, where simultaneous optimization of classification and regression tasks can better demonstrate the overall effectiveness of the model. This structure leads to a fair and relevant comparison, isolating the model's performance in each setting and clearly showing the advantages of multitask learning over individual tasks.

To evaluate the classification performance of our proposed model as a single-task classifier, we compare it with four state-of-the-art deep learning models: U-Net [57], SegNet [69], Deeplab V3+ [61], and MSTNet [70] (a multilevel spectral–spatial transformer network), each known for its strong performance in remote sensing applications.

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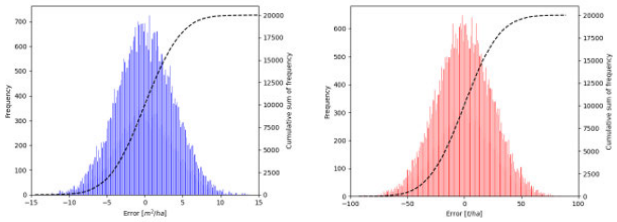
TABLE III

PERFORMANCE EVALUATION FOR ALL TEN DIFFERENT CONTINUOUS VARIABLES. TO ENSURE THAT THE REPORTED PERFORMANCE METRICS ARE ROBUST AND REPRESENT THE MODEL'S TRUE CAPABILITY, WE HAVE CONDUCTED TEN TRIALS/SEEDS OF THE EXPERIMENT TO GET MIN/MAX/MEAN VALUES

Continuous	RMSE	MAE	R^2
Basal area [m^2/ha]	0.027 Min: 0.024 Max: 0.03	0.019 Min: 0.017 Max: 0.021	0.907 Max: 0.913 Min: 0.889
Mean dbh [cm]	0.030 Min: 0.027 Max: 0.032	0.028 Min: 0.026 Max: 0.031	0.913 Max: 0.927 Min: 0.892
Stem density [1/ha]	0.021 Min: 0.019 Max: 0.025	0.021 Min: 0.018 Max: 0.026	0.945 Max: 0.959 Min: 0.936
Mean height [cm]	0.011 Min: 0.010 Max: 0.013	0.030 Min: 0.029 Max: 0.032	0.927 Max: 0.938 Min: 0.916
Percentage of Pine	0.020 Min: 0.019 Max: 0.022	0.027 Min: 0.026 Max: 0.028	0.919 Max: 0.926 Min: 0.907
Percentage of spruce	0.037 Min: 0.036 Max: 0.038	0.036 Min: 0.035 Max: 0.038	0.926 Max: 0.936 Min: 0.918
Percentage of birch	0.038 Min: 0.037 Max: 0.039	0.036 Min: 0.035 Max: 0. 038	0.949 Max: 0.958 Min: 0.937
Woody biomass	0.026 Min: 0.025 Max: 0.027	0.031 Min: 0.030 Max: 0.032	0.951 Max: 0.964 Min: 0.942
Leaf area index	0.033 Min: 0.032 Max: 0.035	0.028 Min: 0.027 Max: 0.029	0.916 Max: 0.927 Min: 0.908
Effective leaf area index	0.028 Min: 0.027 Max: 0.030	0.030 Min: 0.028 Max: 0.031	0.928 Max: 0.931 Min: 0.927

While U-Net, SegNet, and Deeplab V3+ rely on convolutional architectures, MSTNet introduces a transformer-based structure specifically designed to enhance spectral and spatial representation in hyperspectral images. The proposed model demonstrates superior performance in OA, precision, and recall metrics compared to these methods, showcasing its robustness and consistency across ten trials, where it achieves higher average values and lower variability. For regression tasks, we compare our model as a single-task regressor against advanced continuous-variable estimation methods, namely, GPR [71], gradient boosting regressor (GBR) [72], deep Gaussian processes (DGP) [73], and deep Gaussian mixture models (DGMMs) [74]. These models represent a diverse set of approaches, including traditional machine learning (GPR and GBR) and advanced deep learning techniques (DGP and DGMM). The proposed model achieves lower RMSE and MAE values, indicating its capacity for precise and reliable continuous-variable estimation.

Finally, for the multitask learning approach, we compare our proposed model against two methodologies, the Mõttus model and the Pham model, which use the same dataset for multitask learning on classification and regression. By incorporating both classification and regression tasks simultaneously, our proposed model achieves the highest values in OA, precision, and recall for classification and the lowest RMSE and MAE



(a) (b)

Fig. 12. Prediction error histogram for selected continuous forest variables. (a) Mean tree height. (b) Woody biomass.

TABLE IV

COMPARISON WITH STATE-OF-THE-ART METHODS: THE BEST RESULTS FOR EACH SECTION (ONLY CLASSIFICATION, ONLY REGRESSION, AND BOTH, I.E., MULTITASKING) ARE IN BOLD, AND THE SECOND-BEST ARE UNDERLINED (HIGHER VALUES OF OA, PRECISION, AND RECALL INDICATE BETTER PERFORMANCE; LOWER VALUES OF RMSE AND MAE ARE BETTER). OUR MODEL SHOWS HIGHER AVERAGE PERFORMANCE AND LOWER VARIABILITY ACROSS TEN TRIALS

Methods	**Classification**			**Regression**	
	OA (%)	Precision	Recall	RMSE	MAE
U-Net	86.12 Max: 88.63 Min: 83.97	80.81 Max: 83.27 Min: 78.19	79.42 Max: 82.17 Min: 78.01	X	X
SegNet	90.87 Max: 92.13 Min: 88.92	87.46 Max: 90.01 Min: 85.92	87.01 Max: 89.98 Min: 86.07	X	X
Deeplab V3+	89.18 Max: 91.01 Min: 88.96	84.64 Max: 86.78 Min: 83.12	85.91 Max: 87.21 Min: 83.84	X	X
MSTNet	**93.83** Max: 94.11 Min:90.04	<u>92.98</u> Max: 93.28 Min: 91.26	<u>92.14</u> Max: 93.27 Min: 90.93	X	X
Proposed as Single-task (Classification)	<u>93.51</u> Max: 93.92 Min: 92.27	**93.01** Max: 93.88 Min:91.84	**92.48** Max: 92.79 Min: 91.22	X	X
GPR	X	X	X	0.061 Min: 0.058 Max: 0.063	0.067 Min: 0.062 Max:0.071
GBR	X	X	X	0.055 Min: 0.050 Max: 0.057	0.059 Min: 0.054 Max: 0.063
DGP	X	X	X	0.041 Min: 0.037 Max: 0.044	0.039 Min: 0.034 Max: 0.043
DGMM	X	X	X	**0.031** Min: 0.028 Max: 0.033	<u>0.028</u> Min: 0.023 Max: 0.32
Proposed as Single-task (Regression)	X	X	X	<u>0.032</u> Min: 0.030 Max: 0.033	**0.025** Min: 0.022 Max: 0.027
Mõttus Model	84.71 Max: 86.37 Min: 82.78	81.08 Max: 83.59 Min:80.01	82.15 Max: 84.58 Min: 80.95	0.087 Min: 0.085 Max: 0.089	0.082 Min: 0.081 Max: 0.084
Pham Model	82.43 Max: 83.96 Min: 81.01	79.18 Max: 81.48 Min: 78.92	80.64 Max: 82.37 Min: 79.12	0.095 Min: 0.094 Max: 0.096	0.094 Min: 0.093 Max: 0.096
Proposed as Multitask	**95.92** Max: 96.38 Min: 94.01	**94.66** Max: 94.72 Min: 94.50	**94.12** Max: 94.83 Min: 94.03	**0.027** Min: 0.024 Max: 0.003	**0.019** Min: 0.017 Max: 0.021

for regression. This result not only highlights the strength of the model in multitask learning but also demonstrates the significant advantage multitasking offers in performance over isolated single-task learning. This comprehensive comparison underscores the value of multitask learning in achieving consistently high performance across both classification and regression tasks.

D. Ablation Study

This section presents an ablation study to systematically assess the contributions of individual components in our proposed framework and their impact on the overall performance. While we initially considered a broad range of evaluation metrics for both classification and regression

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(see Tables II and III), we opted to report a subset (OA, precision, and recall for classification and RMSE and MAE for regression) in Tables IV–VII. This choice was driven by two key considerations: First, we aimed to present a concise yet comprehensive view of model performance, where the selected metrics capture both the predictive accuracy and the balance between precision and recall for classification, as well as the error magnitude and variability for regression. For classification, OA, precision, and recall were chosen because they are widely interpretable and offer a balanced assessment of model performance across classes. For regression, RMSE and MAE provide complementary insights into error, with RMSE emphasizing larger errors and MAE offering a straightforward interpretation of average error. Second, by focusing on these core metrics, we reduced computational overhead, allowing us to allocate more resources and attention to other aspects of the analysis in this article. The remaining metrics, while informative, were generally aligned with the insights provided by the selected subset, and instead of adding significant distinct information, redundancy can be introduced.

1) Effect of Different Layers in the Network: Table V illustrates the performance evaluation of the impact of processing modules of the proposed network. The bolded values denote the highest metrics achieved among the methods. With the addition of different modules to the baseline, there is an increment in OA, precision, and recall, accompanied by reductions in RMSE and MAE. In particular, adding a multitask loss layer yields the highest precision values and the lowest RMSE, emphasizing its significance in the architecture. Multitask loss enables simultaneous optimization of multiple objectives, thus enhancing model performance by addressing data imbalances and promoting feature learning across related tasks. The channel pooling layer also gives a significant result since the dataset is huge, and using the channel pooling layer reduces dimensionality, thus capturing important features across channels. This ablation study demonstrates the effect of different modules in a network, which helps in understanding how information is transformed throughout the network, aiding in model interpretation, optimization, and architectural improvements.

2) Effect of the Skip, BN, and GAP Layer: The impact of skip, BN, and GAP layers is studied through training loss, assessing how the proposed model converges with and without these layers. The analysis, as shown in Fig. 13, reveals that the combined utilization of all three options notably enhances the convergence of the training loss curve compared to their individual usage. These layers are recognized as effective regularizers for neural networks and are widely embraced by the deep learning community for enhancing model performance and generalization. Skip layers aid in the training of deep networks and reducing vanishing gradient problems, while batch normalization ensures stable and accelerated training by normalizing activations within each mini-batch, improving convergence and regularization. The global average-pooling layer condenses feature maps into a single vector, reducing model complexity and overfitting while preserving spatial information. The skip layer shows its dominance over BN and GAP by showing a significant contribution to the error reduction for this dataset. Thus, including all three layers is essential for accelerating the learning process by effectively reducing the training error.

3) Effect of Cost-Sensitive and Focal Loss: In our proposed approach, we have incorporated cost-sensitive learning and focal loss to tackle class imbalance issues effectively. For cost-sensitive learning, the inverse median frequency method was applied to calculate class weights, which applies strict consequences on errors from minority class samples compared to majority class samples. This weighting scheme was also utilized for the CE term in focal loss. It can be observed from Table VI that the focal loss without the class weights for

TABLE V

EFFECT OF DIFFERENT LAYERS IN THE NETWORK. R: RESNET; A: ATROUS CONVOLUTION; T: ATTENTION NETWORK; M: MULTITASK LOSS; AND C: CHANNEL POOLING

Methods	Classification			Regression	
	OA (%)	Precision	Recall	RMSE	MAE
Baseline	79.12 Max: 79.87 Min: 78.92	82.03 Max: 82.78 Min: 81.67	80.12 Max: 81.02 Min: 79.92	0.091 Min: 0.090 Max: 0.092	0.103 Min: 0.102 Max: 0.105
Baseline + R	84.18 Max: 84.76 Min: 83.92	84.79 Max: 85.28 Min: 84.12	85.18 Max: 85.93 Min: 84.63	0.078 Min: 0.077 Max: 0.079	0.082 Min: 0.081 Max: 0.084
Baseline + RA	87.37 Max: 87.85 Min: 87.01	86.43 Max: 87.12 Min: 86.09	86.01 Max: 86.92 Min: 85.38	0.057 Min: 0.057 Max: 0.058	0.061 Min:0.060 Max: 0.062
Baseline + RAT	90.49 Max: 91.13 Min: 89.93	89.78 Max: 90.34 Min: 89.21	89.37 Max: 90.02 Min: 89.11	0.042 Min: 0.041 Max: 0.043	0.047 Min: 0.045 Max: 0.048
Baseline + RATM	91.87 Max: 92.34 Min: 91.27	**95.02** **Max: 95.78** **Min: 94.82**	90.64 Max: 91.27 Min: 90.92	**0.027** **Min: 0.026** **Max: 0.028**	0.032 Min: 0.031 Max: 0.033
Baseline + ATC	91.13 Max: 91.89 Min: 90.94	91.34 Max: 91.93 Min: 90.72	90.98 Max: 91.34 Min: 90.47	0.033 Min: 0.032 Max: 0.034	0.030 Min: 0.029 Max: 0.031
Proposed	**95.92** **Max: 96.38** **Min: 94.01**	94.66 Max: 94.72 Min: 94.50	**94.12** **Max: 94.83** **Min: 94.03**	0.027 Min: 0.024 Max: 0.003	**0.019** **Min: 0.017** **Max: 0.021**

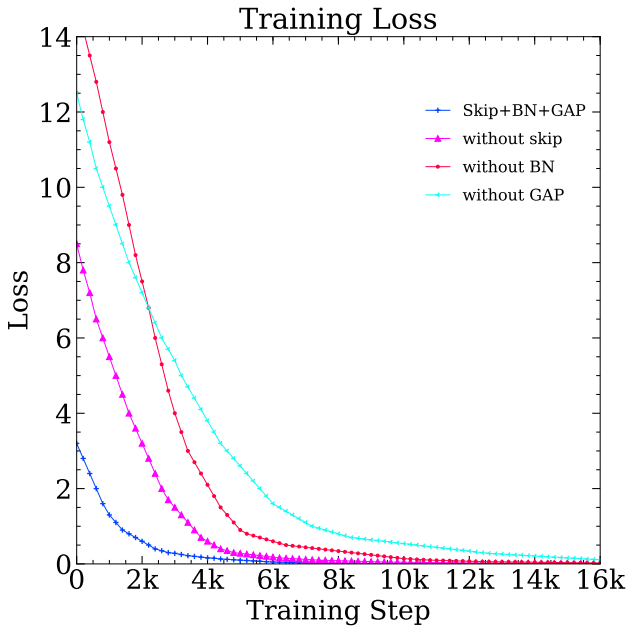


Fig. 13. Effect of skip connections, BN, and GAP layer on the proposed method.

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TABLE VI
EFFECT OF COST-SENSITIVE AND FOCAL LOSS WITH AND WITHOUT CLASS WEIGHTS ON CLASSIFICATION AND REGRESSION EVALUATION METRICS

Methods	Classification			Regression	
	OA	Precision	Recall	RMSE	MAE
Cost-Sensitive	87.23	85.26	86.81	0.073	0.078
Focal Loss without class weights	80.72	79.10	80.23	0.093	0.098
Focal Loss with class weights	**90.09**	**89.63**	**89.12**	**0.051**	**0.057**

the CE loss performs poorly compared to the cost-sensitive learning approach. Therefore, we have incorporated class weights into the focal loss to address data imbalance in our study, ensuring that the model is not biased toward the majority classes.

4) Evaluation of Complexity in the Proposed Multitask Model: We conducted a comparative analysis with state-of-the-art deep learning methods on our dataset to evaluate the tradeoff between performance and complexity in our proposed multitask model. Table VII summarizes key metrics, including model parameters, floating-point operations (FLOPs), training epoch duration, and OA. These evaluations were performed for the classification task, the multitask baseline models (Mõttus model and the Pham model), and the proposed multitask model. We did not extend these evaluations to include state-of-the-art regression models because the complexity metrics typically used to evaluate classification tasks, such as FLOPs, parameter counts, and training time, are less commonly applied to regression models, where predictive accuracy is often the primary concern. Furthermore, including these metrics for regression models may not provide meaningful comparisons due to differences in model structure and evaluation standards. Therefore, we focused on metrics that are more standardized and relevant to the models and tasks under investigation.

Among the models evaluated, Deeplab V3+ shows the highest complexity, characterized by a large parameter count and extensive training time required to achieve satisfactory results. In contrast, the U-Net model, despite its simpler architecture with fewer parameters and shorter training time, demonstrated average performance accuracy. Furthermore, while the MSTNet showcased the second-highest accuracy (underlined in Table IV), its substantial parameter size and increased FLOPs resulted in prolonged training times, making it less practical for large-scale hyperspectral datasets. On the other hand, our proposed multitask model has the lowest parameter count as it utilizes shared encoder layers and an ASPP module with varied dilation rates. This significantly reduces the parameter count compared to its single-task counterpart and other state-of-the-art methods. This architectural choice not only minimized training time but also enhanced efficiency in capturing multiscale contextual information without the need for additional parameters. To provide a comprehensive evaluation, we report the total computing time spent on evaluating each method, considering the initial conditions and variations in computational resources. Our method's total evaluation time was 42.8 h (training time per epoch $\times$ number of epoch), while the second-best OA performance model, MSTNet's total evaluation time, was 88.95 h. This indicates that the proposed multitask model emerged as a lightweight yet effective solution, striking a balance between model complexity, training efficiency, and performance accuracy suitable for large-scale hyperspectral data analysis.

TABLE VII
EVALUATION OF COMPLEXITY IN THE PROPOSED MULTITASK MODEL COMPARED TO STATE-OF-THE-ART METHODS FOR CLASSIFICATION TASK

Methods	Number of Parameters (M)	FLOPs (G)	Training Time per epoch (sec)	OA (%)
U-Net	<u>28.93</u>	112.67	**112.83**	86.12
SegNet	51.44	160.41	1219.48	90.87
Deeplab V3+	53.6	82.7	305.98	89.18
MSTNet	54.28	102.46	2134.92	<u>93.83</u>
Mõttus Model	37.82	61.28	<u>183.42</u>	84.71
Pham Model	41.37	**32.48**	247.39	82.43
Proposed as Single-task	37.32	78.14	4593.67	93.51
Proposed as Multitask	**19.68**	<u>41.79</u>	1028.76	**95.92**

E. Effect of Input Patch Size

To thoroughly understand the impact of spatial input patch size on both categorical and continuous predictions, we conducted an ablation study to analyze different patch sizes. This analysis is crucial as the spatial context captured in HSI can significantly affect model performance, particularly in multitask learning, where both classification and regression tasks are involved. Here, we aimed to assess how varying patch sizes affect OA (for classification tasks) and RMSE, as well as R^2 scores (for regression tasks) across the selected variables. For the categorical tasks, as shown in Fig. 14, we observed that the accuracy for the fertility class, soil class, and mean tree species initially increases with patch size, reaching a peak at approximately 39×39 or 45×45 pixels, depending on the class. Beyond this optimal range, performance begins to decline, suggesting that larger patches may introduce redundant or irrelevant information, which can interfere with the classification performance. This effect aligns with our expectation that while larger patches may capture more spatial context, they can also add noise or reduce the model's ability to focus on relevant features, thus reducing classification accuracy. In terms of continuous variables, as seen in Table VIII, we evaluated the effect of patch size on basal area, mean height, and woody biomass out of the ten continuous variables. This selection was made to balance computational efficiency with the need to provide a representative analysis. For these regression tasks, we also observed that mid-range patch sizes, particularly 39×39 and 45×45, provide the best balance in terms of RMSE and R^2 scores, which confirms the results of Mõttus et al. [24], using a different model. Thus, both categorical and continuous tasks benefit from an intermediate patch size, which appears to provide enough spatial context without overwhelming the model with unnecessary details. These findings demonstrate that choosing an optimal patch

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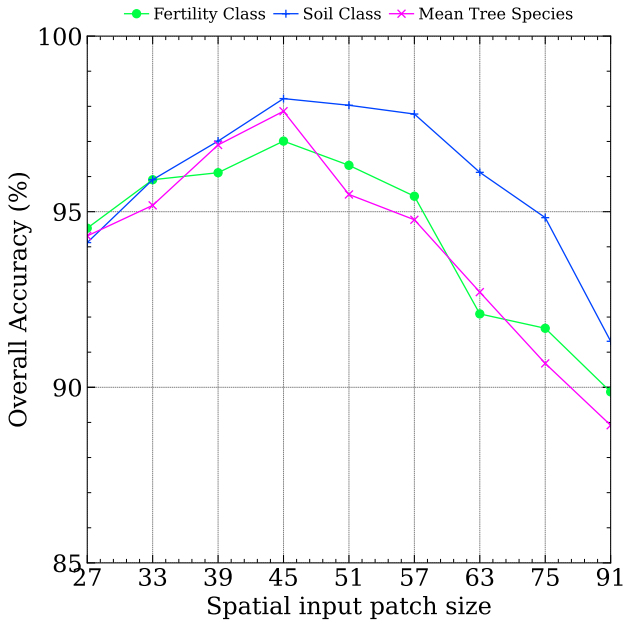


Fig. 14. Effect of input patch size on categorical classes.

TABLE VIII
EFFECT OF INPUT PATCH SIZE ON CONTINUOUS CLASSES

Input Patch Size	Basal area [m2/ha]		Mean height [cm]		Woody biomass	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
27x27	0.043	0.851	0.041	0.896	0.054	0.912
33x33	0.039	0.873	0.028	0.904	0.048	0.937
39x39	0.032	0.897	0.019	0.917	0.033	0.951
45x45	**0.028**	**0.904**	**0.011**	**0.929**	**0.025**	**0.958**
51x51	0.064	0.882	0.038	0.904	0.043	0.947
57x57	0.091	0.864	0.069	0.881	0.071	0.936
63x63	1.081	0.861	0.089	0.858	0.094	0.921
75x75	1.973	0.832	1.010	0.822	1.192	0.904
91x91	3.932	0.811	1.794	0.804	1.623	0.893

size is essential to maximizing both classification accuracy and regression precision in multitask hyperspectral image analysis.

V. DISCUSSION

This study introduces a multitask deep learning model that simultaneously addresses classification and regression tasks within the domain of HSI for forestry applications. Traditional approaches in hyperspectral analysis largely rely on single-task learning, which limits computational efficiency and scalability when addressing complex, multidimensional characteristics of forest data. In contrast, our model efficiently predicts both categorical and continuous forest variables, optimizing resource use and providing a richer, integrated understanding of the data by utilizing shared spectral and spatial information. This multitask approach improves both predictive accuracy and interpretability, especially in large-scale, high-dimensional datasets like TAIGA.

Key findings indicate that the proposed model significantly outperforms single-task frameworks in both classification and regression tasks. Achieving an OA of 95.92% on classification tasks, the model outperforms traditional single-task methods in both adaptability and computational efficiency. Furthermore, regression tasks demonstrate significantly lower error margins attributed to the effective integration of skip connections, batch normalization, and global average-pooling layers. These architectural choices have also played a critical role in maintaining model stability, preventing overfitting—a common risk in high-dimensional hyperspectral data. By enhancing mean class accuracy, especially in minority classes, the model addresses class imbalance robustly and represents a balanced solution for complex, mixed-variable forestry datasets.

Despite these strengths, several limitations should be acknowledged. Computational demand remains high due to the model's complex structure, which balances shared and task-specific learning while managing high-dimensional hyperspectral data. Although this multitask approach is more efficient than running separate models for classification and regression, further optimization, such as model compression or parameter pruning, could reduce computational costs, enhancing its suitability for resource-constrained environments. The inherent opacity of multitask models presents another challenge, while attention mechanisms improve interpretability; further work in model explainability is necessary. Techniques like layer-wise relevance propagation or feature visualization could illuminate the internal decision-making processes, especially valuable when deploying such models in environmental monitoring and ecological applications where transparency is essential.

Overfitting remains a potential risk due to the model's high parameter count and multitask balancing requirements. Although performance across trials was consistent, indicating generalizability within the TAIGA dataset, this dataset, while extensive, cannot capture the full variability of forest ecosystems. For instance, forests with differing soil compositions, biodiversity, or spectral properties—even within the same latitude—could pose significant challenges for the model. These differences may limit its applicability to other regions, highlighting the need for additional strategies to enhance generalization. Future work could explore domain adaptation techniques or semisupervised learning to incorporate ecological variability into training.

To further assess generalization capabilities, testing the model on datasets like Toulouse [75], despite potential task and label differences, could provide valuable insights. However, challenges such as variations in sensor calibration and spectral ranges must be carefully considered to ensure fair comparisons. Testing this model on alternative datasets for classification or regression tasks could also help evaluate its robustness against state-of-the-art approaches in broader contexts. By addressing these limitations, we aim to build a more adaptable framework capable of supporting diverse forestry and ecological applications.

Thus, this research advances multitask learning in hyperspectral forestry analysis by addressing class imbalance, optimizing attention mechanisms, and prioritizing computational efficiency. It establishes a robust foundation for simultaneous classification and regression tasks in complex forestry datasets like TAIGA. While further refinements in model explainability and adaptability are necessary, this framework positions multitask learning as a valuable tool in remote sensing, promising broader applications in environmental monitoring as the field continues to evolve.

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VI. Conclusion

In this article, we proposed a novel multitask deep learning model for concurrent classification and regression of hyperspectral images. We have considered the TAIGA dataset, a larger hyperspectral dataset that consists of 13 forest variables, among which are three categorical (with different subclasses) and ten continuous variables (biophysical parameters). We considered sharing encoder and task-specific decoder architecture in the proposed framework. In addition, a dilated convolutional block called dense ASPP was considered to capture multiscale contextual information effectively. Furthermore, considering the importance of both spectral and spatial information, we implemented a spectral–spatial attention block to enable selective information processing where task-specific features were prioritized, improving generalization. We also integrated cost-sensitive learning and focal loss techniques to address the class imbalance issue and achieve task balancing in the proposed multitask framework. We aimed to improve model performance and efficiency across various tasks by computing multitask loss and optimizing the model parameters within this framework. Extensive experiments conducted on a large hyperspectral dataset showcase the superiority of the proposed method compared to both classical and state-of-the-art models. Ablation studies further validate the effectiveness of different parameters used for this work. Further work is being taken up where the model can be tuned to learn more tasks on different datasets for HSI classification and reduce the computation time by an iterative process, thus increasing algorithm efficiency.

Acknowledgment

The work of Koushikey Chhapariya was supported by the Indo-French Centre for Promotion of Advanced Research (IFCPAR/CEFIPRA) through the Raman–Charpak Fellowship Program 2022.

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